ImageClassification

Dimitrios Pakakis

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1 Introduction

In this project, we use Principal Component Analysis (PCA) to reduce the dimensions of our data and test how well we can reconstruct the original images. We also use the K-Means algorithm to group the data automatically, and we build a Nearest Centroid Classifier to categorize new images.

Our main goal is to understand how reducing dimensions affects our ability to separate different clothing classes, and to check how much information is lost during this compression. Finally, we compare two ways of measuring distance, the Euclidean distance versus the Mahalanobis distance, to see which one works best.

2 Dataset

Fashion-MNIST is a dataset of Zalando's article images, consisting of a training set of 60,000 examples and a test set of 10,000 examples. Each example is a 28x28 grayscale image, associated with a label from 10 classes. Each image is 28 pixels in height and 28 pixels in width, for a total of 784 pixels in total. Each pixel has a single pixel-value associated with it, indicating the lightness or darkness of that pixel, with higher numbers meaning darker. This pixel-value is an integer between 0 and 255.

3 Evaluation Metrics

For the evaluation i will use:

- Accuracy: Number of correct predictions
- Macro Precision: average for all the classes of how often the model predictions are false (FP)
- Adjusted Rand Index (ARI) : To evaluate clustering similarity against ground truth. It computes a similarity score between assignments and true labels
- Confusion Matrix: It compares the predictions made by the model with the actual results and shows where the model was right or wrong.

For the accuracy,precision and ARI 100% means perfect performance.

4 Algorithms implementation

We implemented the algorithms from scrathc from fundamental linear algebra primitives to ensure a deep understanding of their mathematical mechanics. Below is a detailed breakdown of the internal logic for each algorithm:

4.1 Principal Component Analysis (PCA)

PCA maximizes the projected variance by solving a constrained optimization problem, proving that the optimal projection axes are the eigenvectors of the data's covariance matrix.

1. **Mean Centering & Covariance:** The dataset is centered by subtracting the empirical mean μ. Then, the covariance matrix Σ is computed to capture feature correlations.
2. **Eigendecomposition:** We extract the eigenvalues and eigenvectors of Σ. By selecting the top K eigenvectors corresponding to the largest eigenvalues, we form the projection matrix U.
3. **Projection & Reconstruction:** Data is projected into the lower-dimensional space via $Z = X_{centered}U$. To

evaluate information retention, we reconstruct the data back to the original space using $X_{recon} = ZU^T + \mu$.

4.2 K-Means Clustering

The K-Means algorithm treats clustering as an alternating minimization problem to reduce the total distortion (inertia) of the groups.

1. **Initialization:** K random data points are selected from the dataset to serve as the initial centroids.

2. **Alternating Optimization:** The algorithm iteratively performs two steps:

   - *Assignment:* Each data point is assigned to the nearest centroid based on its Euclidean distance.
   - *Update:* Each centroid is recalculated as the spatial mean of all points assigned to its cluster.

3. **Convergence:** This process repeats until the centroids' positions stabilize, indicating that a local minimum has been reached.

4.3 Nearest Centroid Classifier (with Subclass Support)

To handle the non-linear boundaries of the clothing categories, we developed an adaptable Nearest Centroid classifier.

1. **Subclass Partitioning:** To handle multi-modal distributions (e.g., short-sleeve vs. long-sleeve shirts), each target class can be dynamically split into distinct subclasses using K-Means during training.

2. **Calculation:** For each subclass, we calculate its mean vector μ , covariance matrix Σ and inverse covariance matrix Σ^{-1}.

3. **Inference:** New samples are classified by finding the closest subclass center. We evaluate two distance metrics: the standard Euclidean distance and the Mahalanobis distance ($D_M^2 = (x-\mu)^T \Sigma^{-1} (x-\mu)$), which leverages the inverse covariance matrices to account for the spread of the data.

5 Dimensionality Reduction with PCA

We restrict the dataset to three specific garment categories: Class 0 (T-shirt/top), Class 6 (Shirt), and Class 7 (Sneaker), and project them into a 3D subspace using the top three principal components.

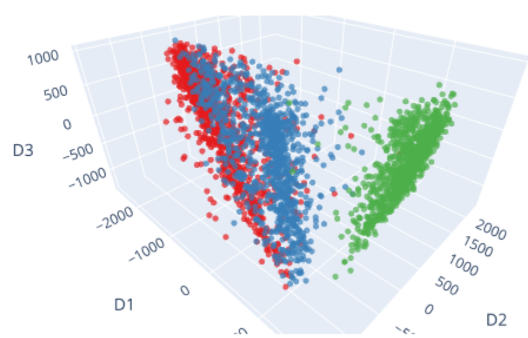

Figure 1: In the figure we can detect that the Class 7 (Sneakers) clearly separates from the other two classes, forming its own distinct group. This happens because the shape of a shoe is completely different from upper-body clothing, so their active pixels do not overlap. On the other hand, Class 0 (T-shirts) and Class 6 (Shirts) are heavily mixed together. Since they share almost the exact same shape (sleeves, neck, and torso), their pixel patterns are very similar. A behaviour that we observed and in the previous analysis.

6 Clustering with K-Means

We executed a 3-cluster K-Means algorithm directly on the 3D projected data. Figure 2 and 3 evaluates the cluster partitions against the actual category labels.

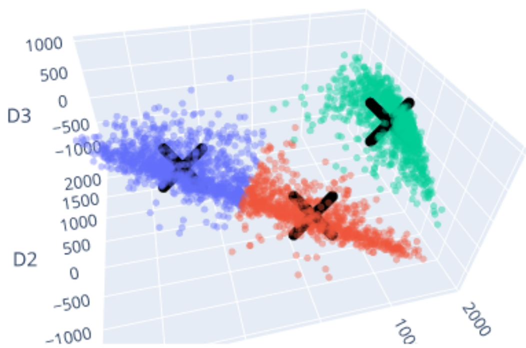

Figure 2: Clusters from K-Means, the X for eahc class is the center.

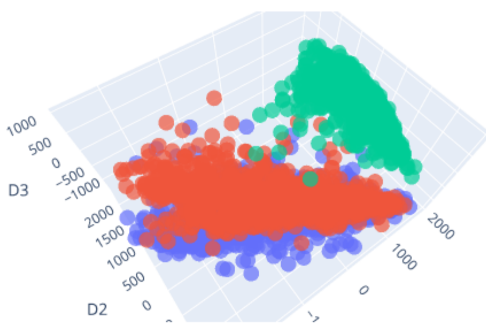

Figure 3: The ground truth.

From the visual comparison we can see a difference between the groups K-Means created and the actual clothing categories. K-Means assumes that groups form perfectly round circles. Because T-shirts (Class 0) and Shirts (Class 6) are mixed together in one big cloud, the algorithm simply cuts this cloud in half, completely missing the real categories. This poor grouping is confirmed by a very low ARI score of 0.4961.

7 Reconstruction Analysis

To inspect the information preserved in the latent subspaces, the cluster centroids discovered in lower dimensions were mapped back into the original 784-dimensional pixel space using the PCA reconstruction function.

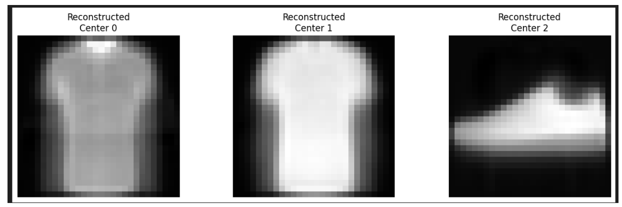

Figure 4: We can see that the reconstruction gave us good, although blurry, shapes for each category. The outlines of the clothes are clear, which confirms again that classes 0 and 6 are very similar.

8 Nearest Centroid Classifier Performance Evaluation

We evaluated the custom Nearest Centroid Classifier across all 10 classes of the Fashion-MNIST dataset, projected into a 10-dimensional PCA subspace. We comprehensively tested two distance metrics: Euclidean distance and Mahalanobis distance, under two distinct configurations: 1 global archetype per class and 2 distinct subclasses per category.

8.1 Mahalanobis Distance and 1 subclass

Accuracy : 75% , Macro precision : 75.7%

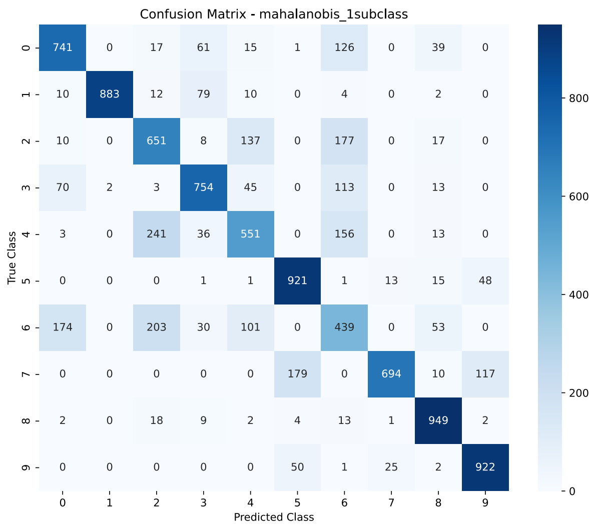

Figure 5: Confusion Matrix.

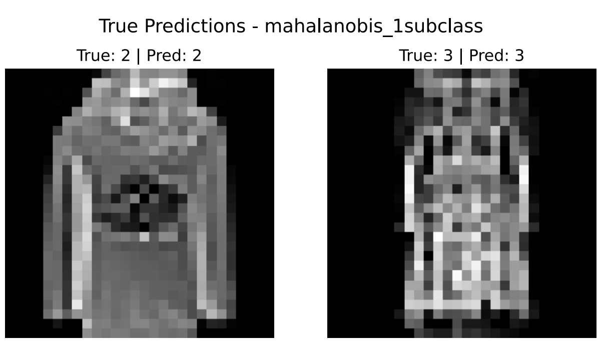

Figure 6: True Predictions.

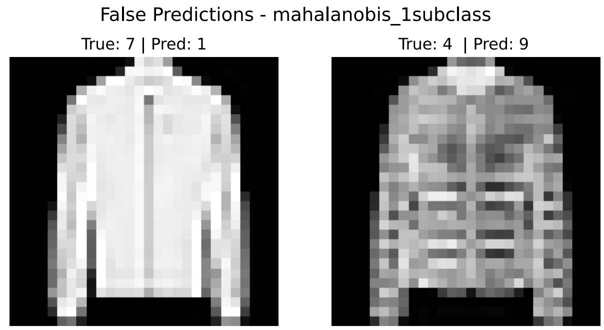

Figure 7: False Predictions.

8.2 Euclidean Distance and 1 subclass

Accuracy : 65.6% , Macro precision : 65.9%

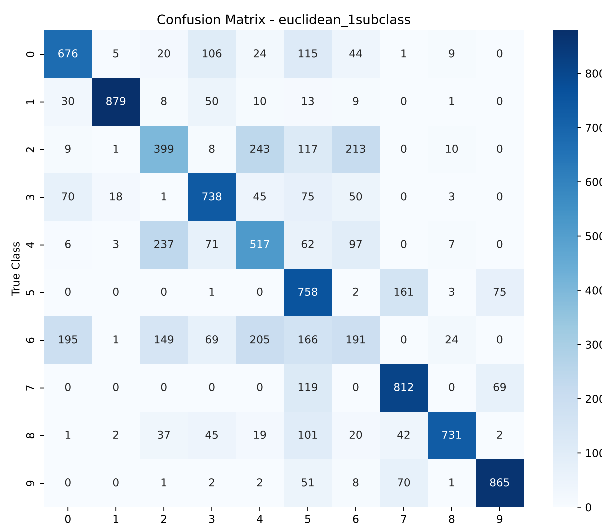

Figure 8: Confusion Matrix.

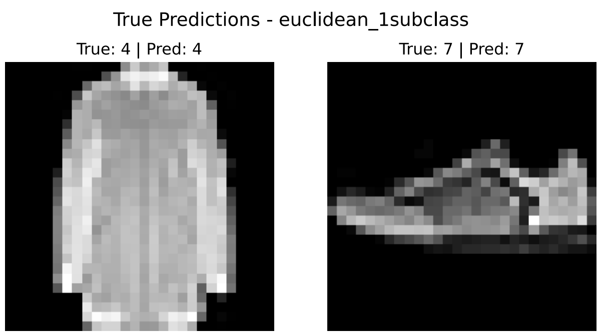

Figure 9: True Predictions.

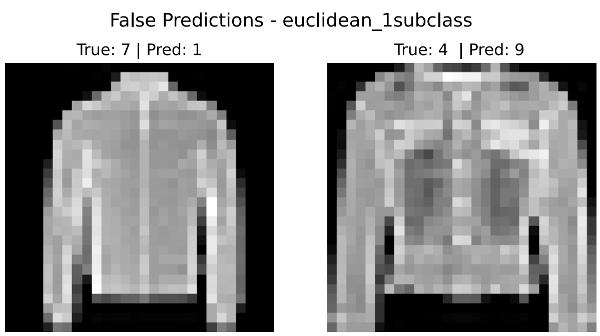

Figure 10: False Predictions.

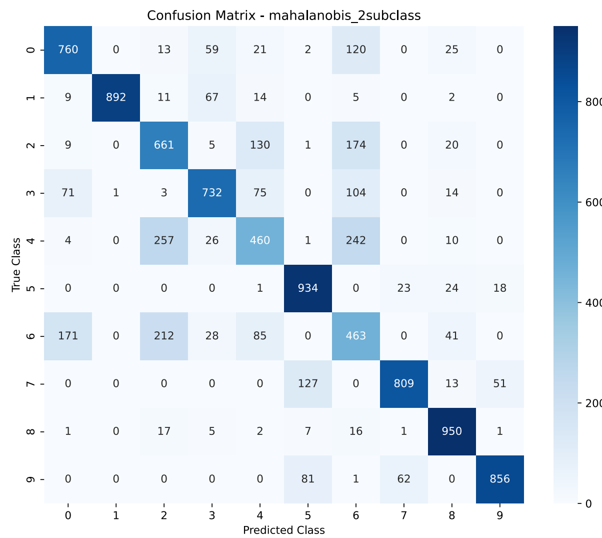

Figure 11: Confusion Matrix.

8.3 Mahalanobis Distance and 2 subclass

Accuracy : 75.1% , Macro precision : 75.9%

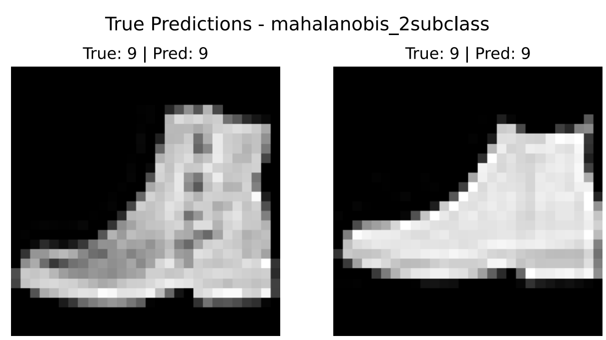

Figure 12: True Predictions.

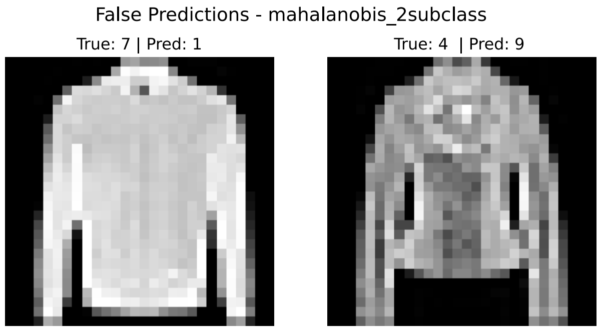

Figure 13: False Predictions.

8.4 Euclidean Distance and 2 subclass

Accuracy : 69% , Macro precision : 69.1%

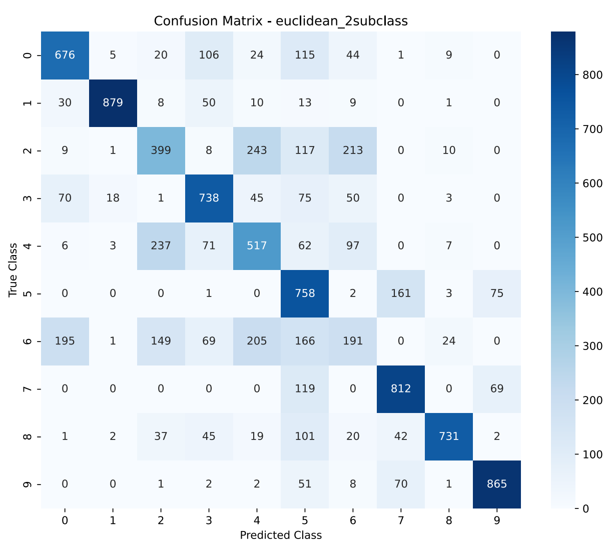

Figure 14: Confusion Matrix.

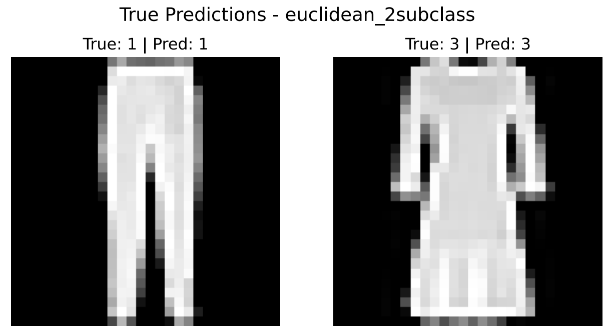

Figure 15: True Predictions.

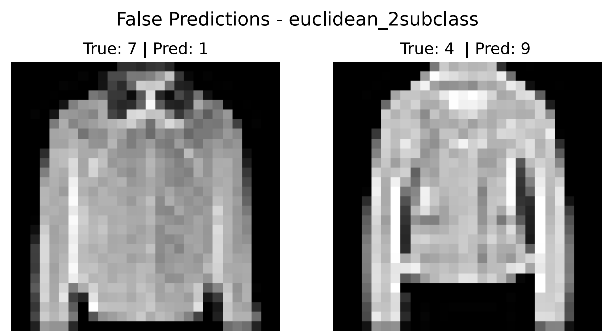

Figure 16: False Predictions.

9 Comments

Looking at the final results and metrics, we can draw two major conclusions regarding the geometry of our data and the classification methods we used:

1. The Superiority of the Mahalanobis Distance The Mahalanobis distance outperformed the Euclidean approach by a large margin. Euclidean distance assumes that the data is shaped like a perfect sphere, giving equal importance to all directions. However, after applying PCA, our data is stretched because some principal components have much larger variances than others. Euclidean distance gets easily confused by this stretched shape. Mahalanobis distance fixes this by using the covariance matrix. It understands the actual "stretched" shape and spread of the data. Instead of just measuring a straight line, it measures the distance

based on the statistical distribution of the data, which leads to much more accurate decision boundaries.

2. The Problem of Identical Clothes and the Impact of Subclasses When we used K-Means to split each clothing category into 2 subclasses, we only saw a minimal increase in accuracy. We understand that the main problem is not that each class contains many different hidden sub-styles (like long-sleeve vs. short-sleeve shirts). If that were the case, creating subclasses would have caused a jump in accuracy. Instead, the real issue is exactly what we observed in the 3D plots: different categories of clothes simply look too much alike. Classes like T-shirts, Shirts, and Coats share almost the exact same basic outline. Because they are visually identical, their data points overlap heavily in the feature space. While splitting them into subclasses helps a little bit, it cannot fully solve the core problem: the fundamental visual similarity between different types of clothing.